Negotiated Autonomy

How Personal AI Learns Its Scope of Action in Spatial Work

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Abstract

An AI rearranges a workspace and moves a monitor its user expected to remain untouched. Understanding the intended layout did not establish permission for that action. I propose that most spatial grants have a compact scope: visible objects or a region, a permitted operation and a duration. I will test this representation and build a desktop/XR collaborator that negotiates grants through speech and learns defaults for repeated actions. Comparisons with an editable rule panel will measure authorisation effort alongside unwanted intervention. The research will identify where compact spatial permissions work, where they fail and how agreements change through use.

1 Motivation and Proposition

“Improve this workspace, but keep the window workstation” leaves room for disagreement about moving its monitor. “Move these chairs while I am away” explicitly delegates work; “not that one” narrows it. My goal is personal AI that connects conversation, contextual perception and execution through agreements people understand and revise.

H1: under a disclosed session-end expiry default, at least 80% of positive grants and grant replacements in free-form room-editing work fit a fixed schema: visible objects or a region, one permitted operation and expiry. Faithful representation preserves the actions permitted and prohibited throughout the grant, rather than merely filling three fields. Study 1 freezes the coding rules before data collection and tests where this compact representation breaks.

2 Related Work and Research Gap

Reference-Based Manipulation formalises spatial reference through Source, Anchor and Frame [1]. EchoLadder offers reversible scene suggestions and documents how users inspect subtle changes [2]. This project adds standing grants with expiry and a revisable history across sessions, testing whether a compact scope record preserves what people authorise.

Orality [3], Rambler [7] and IntentPrism [8] support developing and inspecting intentions. Authorisation has a further function: “those chairs are awkward” expresses judgement; “move those chairs until I return” grants action. Study 1 classifies these speech acts in spatial work.

Horvitz addresses mixed initiative under uncertainty [4]; Parasuraman et al. distinguish automation types and levels [5]; Amershi et al. guide correction and adaptation [6]. Sensible Agent adapts proactive AR assistance to context [9]. This project tests the adequacy of spatial permission records and the cost of negotiating them, including when rules outperform dialogue.

3 Research Questions

RQ1: How are spatial permissions granted and revised in speech and action, and which grants defeat the proposed compact representation?

RQ2: How do a rule panel, conversational authorisation and conversational authorisation with learned defaults differ in delegation overhead, unwanted actions and task completion?

RQ3: How do standing grants and preferred execution modes change across sessions, and when do authorised action and user expectations diverge?

4 Prototype and Mechanisms

4.1 Representing a grant

The first prototype grounds speech in editor state, using a resettable room and fixed assets; gaze and first-person video are later extensions for perception beyond the editor. For “move chairs on this side of the desk”, Source is the desk, Anchor its edge and Frame the user's viewpoint [1]. A gesture

selects scope; floating models preview edits and labels show operation and expiry. Timestamped records distinguish the scene at authorisation from the scene at execution.

4.2 Commitment and continued work

Suggestions, ghost previews, sandbox edits and commitment provide increasing levels of action. Previews are discarded, sandboxes reset and committed edits retain undo. Background work receives its own time-limited grant. A standing agreement governs later actions, recording who changed its scope and when.

4.3 Learning reusable defaults

For each user and object-operation class, an ordinal estimator learns a default: suggest, preview, sandbox or commit. It uses the history of explicit default-mode choices and corrections; silence supplies no label. A standing grant sets the scope and maximum level, within which defaults govern repeated actions. Scope expansion requires a new grant. A permission history records mode changes and their evidence. Study 2b tests this learner against simply reusing the last explicitly approved mode.

4.4 Interrupting and returning control

A spoken stop or palm gesture cancels queued edits in the indicated scope. Direct manipulation pauses conflicting edits. The interface shows completed changes and unfinished work before renewed instruction resumes execution.

5 Studies and Evaluation

Study 1: Speech acts and coverage. Fourteen participants will redesign workspaces with a Wizard-of-Oz assistant. Before tasks, participants learn that omitted expiry lasts until they close the workspace. Grants remain revocable; temporary absence does not end the session, and cross-session continuation requires an explicit grant. Free-form work precedes exposure to the three-field schema and diagnostic counterexamples.

Before recruitment, I will freeze a codebook: the scope is a finite set of visible object IDs or a drawn region; the operation is move, rotate, resize, add or remove; expiry is a time, session end or a named return event. Excluding an identified object is set subtraction. State-dependent exceptions, dependencies, numerical action budgets and delegated judgement ("make it comfortable") fail unless the original meaning is preserved without additional fields. They cannot be compiled into a chosen subset, split into easier grants or hidden in operation labels.

The denominator includes every positive grant or replacement from free-form work, retaining attached conditions. Pure prohibitions and withdrawals form a separately reported speech-act category. Two coders assess participant-confirmed meaning; unresolved cases stay in the denominator as failures. I will report coverage with participant-level uncertainty and counts for explicit versus default expiry. Targeted counterexamples diagnose failures outside the H1 denominator.

Study 2a: Interface and change frequency. Thirty participants will compare (A) an editable rule panel, (B) conversational authorisation and (C) conversation with learned defaults. Each interface has low-change and high-change blocks: one versus four task-constraint changes requiring scope revision during eight matched editing opportunities. Tasks, permission options and execution controls are shared; rules remain editable. Interface order and task assignment are counterbalanced.

H2 predicts lower authorisation time for the panel at low change frequency and for dialogue at high frequency. The primary contrast is the interface-by-frequency interaction for A versus B. C receives the same initial defaults. Delegation overhead and unwanted actions are primary outcomes, with completion and repair reported alongside participant and task effects.

Study 2b: Learning beyond the last approval. Twenty-four participants will compare C with (D) reusing the last explicitly approved default mode across four sessions. Each user encounters two predefined object-operation classes: four shared calibration choices per class, then 24 opportunities per class per policy (104 events per user). Matched task sequences and policy order are counterbalanced; histories update separately. Both policies share grants, ceilings and correction

controls. Calibration costs and actual explicit mode labels are reported alongside delegation time, unwanted actions and completion in stable and changing contexts.

Measure	Operational definition
Delegation overhead	Authorisation utterances, gestures, panel edits and active time; initial setup and later revisions separated.
Autonomy drift	Executed edits outside the latest explicit grant, coded by detection delay; unnoticed at 120 active-inspection seconds marks delayed drift.
Regret and repair	Unwanted actions, reason-coded undo, and time or edits required to restore intended work.
Reliance and completion	Predicted scope adherence versus observed adherence; user-endorsed completion, latency and checking effort.

Detection time starts when a change is displayed during active inspection, including after return from absence; changes not displayed remain unassessed. Continuous detection time, exploratory undo, later preference changes and unwanted actions within valid grants are coded separately. Grant violations audit grounding and execution. Model claims, tool success and user-goal achievement remain distinct.

Study 3: Agreements across sessions. Twelve participants will use the prototype for six weeks. Weekly event reviews reconstruct permission histories, distinguishing deliberate revision from unrecognised divergence and answering RQ3.

Preliminary Work

Game Fellows provides a bounded companion and a prepared Solo/Fellow comparison with paired seeds, event logs and perceived-control questions; human evaluation is planned. In an image-detection capstone, I reconstructed 900 decisions from frozen outputs and coefficients, exposing class-specific trade-offs hidden by overall improvement. This experience supports reconstructing actions against their contemporaneous permissions.

6 Contributions and Schedule

The thesis will contribute an account of authorisation speech acts, evidence for the coverage and limits of compact spatial grants, and tested mechanisms for reducing delegation effort while tracking unwanted actions.

Period	Work and output
Year 1	Speech-act study; coverage test; desktop/XR prototype.
Year 2	Standing grants and mode learning; technical evaluation.
Year 3	Rule-panel and dialogue comparisons; overhead and action audit.
Year 4	Six-week deployment; synthesis and thesis.

Studies will obtain ethics approval and informed consent, minimise recorded data, and support withdrawal and deletion.

7 Fit and First Collaboration

Can Liu's work connects spatial reference [1], reversible scene design [2] and spoken thought [3]. My architectural training and CAVE installation experience provide practical grounding for testing that connection. In the first six months, I propose building a spatial permission mock-up and conducting the Wizard-of-Oz speech-act study, establishing which grants a compact representation preserves before implementing learned defaults.

References

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